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Supplemental information

Electrochemical system analysis

from impedance data to system identification

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Note S1: Loewner framework

The following section presents the steps required to calculate the Loewner framework. These steps are derived from [28]. Further information on the procedure can be found in the literature[39, 40, 36].
To calculate the state space matrices with the Loewner framework, the measurement data is first divided into two sets. Different methods can be used for this, whereby the alternate split

$$\{s, G(s) | 1 \leq l \leq 2n\} \rightsquigarrow \begin{matrix} \text{right data:} & \left\{\left(\psi_j, G\left(\psi_j\right)\right), j = 1, \dots, n\right\} \\ \text{left data:} & \left\{\left(\gamma_i, G(\gamma_i)\right), i = 1, \dots, n\right\} \end{matrix} \quad \text{(Equation S 1)}$$

is used in this work. ψ and γ describe the evaluated points of the Laplace domain or, more specifically, the measured frequency points in the context of impedances.
In the subsequent step, the values of the transfer function are transferred to the vectors of the right and left sets of the transfer functions $\mathbb{W}$ and $\mathbb{V}$ as follows

$$\mathbb{V} = \begin{bmatrix} G(\gamma_1) \\ G(\gamma_2) \\ \vdots \\ G(\gamma_n) \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}, \quad \mathbb{W} = \begin{bmatrix} G(\psi_1) \\ G(\psi_2) \\ \vdots \\ G(\psi_n) \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}, \quad \text{(Equation S 2)}$$

The split data is now converted to a matrix format using the Loewner matrix $\mathbb{L}$ and the shifted Loewner matrix $\mathbb{L}_s$:

$$\mathbb{L} = \begin{bmatrix} \frac{\boldsymbol{v_1 - w_1}}{\boldsymbol{\gamma_1 - \psi_1}} & \cdots & \frac{\boldsymbol{v_1 - w_n}}{\boldsymbol{\gamma_1 - \psi_n}} \\ \vdots & \ddots & \vdots \\ \frac{\boldsymbol{v_n - w_1}}{\boldsymbol{\gamma_n - \psi_1}} & \cdots & \frac{\boldsymbol{v_n - w_n}}{\boldsymbol{\gamma_n - \psi_n}} \end{bmatrix}, \quad \text{(Equation S 3)}$$

$$\mathbb{L}_{\mathbf{s}} = \begin{bmatrix} \frac{\boldsymbol{\gamma_1 v_1 - \psi_1 w_1}}{\boldsymbol{\gamma_1 - \psi_1}} & \cdots & \frac{\boldsymbol{\gamma_1 v_1 - \psi_n w_n}}{\boldsymbol{\gamma_1 - \psi_n}} \\ \vdots & \ddots & \vdots \\ \frac{\boldsymbol{\gamma_n w_n - \psi_1 w_1}}{\boldsymbol{\gamma_n - \psi_1}} & \cdots & \frac{\boldsymbol{\gamma_n v_n - \psi_n w_n}}{\boldsymbol{\gamma_n - \psi_n}} \end{bmatrix}. \quad \text{(Equation S 4)}$$

These two matrices are also referred to as Loewner pencils, as they can be employed to reconstruct the space matrices

$$\boldsymbol{E}_{\mathrm{L}} = -\mathbb{L}, \quad \boldsymbol{A}_{\mathrm{L}} = -\mathbb{L}_{\mathrm{s}}, \quad \boldsymbol{B}_{\mathrm{L}} = \mathbb{V}, \quad C_{\mathrm{L}} = \mathbb{W} \quad \text{(Equation S 5)}$$

and thus also the transfer function purely data-driven.

Note S2: Curvature of discrete 2d curves

A detailed explanation of the following steps can be found in [41]. The curvature of a general parametrized 2-dimensional curve $\vec{\gamma}(t)$ can be calculated with

$$\kappa(t) = \frac{\det\left[\frac{d}{dt}\vec{\gamma}(t), \frac{d^2}{dt^2}\vec{\gamma}(t)\right]}{\left\|\frac{d}{dt}\vec{\gamma}(t)\right\|_2^3}. \quad \text{(Equation S 6)}$$

Measured and modelled impedances are typically discrete rather than continuous, requiring an approximation. Here, the curvature is calculated from the reciprocal radius $\boldsymbol{r}$ of the osculating circle spanned by the impedance point $\boldsymbol{Z(\omega_k)}$ and its two adjacent points $\boldsymbol{Z(\omega_{k-1})}$ and $\boldsymbol{Z(\omega_{k+1})}$.

$$\kappa\left(Z(\omega_k)\right) = \frac{1}{r_k} = \frac{2\sin(\alpha_k)}{\|\vec{c_k}\|_2} \quad \text{(Equation S 7)}$$

$\boldsymbol{\alpha}$ is the geometric angle between the two vectors $\overrightarrow{\boldsymbol{Z(\omega_{k-1})}, \boldsymbol{Z(\omega_k)}}$ and $\overrightarrow{\boldsymbol{Z(\omega_k)}, \boldsymbol{Z(\omega_{k+1})}}$, where $\vec{\boldsymbol{c}}$ is defined as the vector spanned by $\overrightarrow{\boldsymbol{Z(\omega_{k-1})}, \boldsymbol{Z(\omega_{k+1})}}$.

Note S3: Observability of states and identifiability of parameters

The state space description of a linear, time-invariant system with n states, p inputs, and q outputs is defined as:

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\,\boldsymbol{x}(t) + \boldsymbol{B}\,\boldsymbol{u}(t) = \underbrace{\begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{bmatrix}}_{n\,x\,n} \boldsymbol{x}(t) + \underbrace{\begin{bmatrix} b_{1,1} & \cdots & b_{1,p} \\ \vdots & \ddots & \vdots \\ b_{n,1} & \cdots & b_{n,p} \end{bmatrix}}_{n\,x\,p} \boldsymbol{u}(t)\,, \quad \text{(Equation S 8)}$$

$$\boldsymbol{y}(t) = \boldsymbol{C}\,\boldsymbol{x}(t) + \boldsymbol{D}\,\boldsymbol{u}(t) = \underbrace{\begin{bmatrix} c_{1,1} & \cdots & b_{1,n} \\ \vdots & \ddots & \vdots \\ c_{q,1} & \cdots & c_{q,n} \end{bmatrix}}_{q\ x\ n} \boldsymbol{x}(t) + \underbrace{\begin{bmatrix} d_{1,1} & \cdots & d_{1,p} \\ \vdots & \ddots & \vdots \\ d_{q,1} & \cdots & d_{q,p} \end{bmatrix}}_{q\ x\ p} \boldsymbol{u}(t) \qquad \text{(Equation S 9)}$$

Observability of states

Kalman's observability criterion states, that an LTI system is observable[2], if and only if the column rank of the observability matrix, defined as

$$\mathbb{O} = \begin{bmatrix} \boldsymbol{C} \\ \boldsymbol{CA} \\ \vdots \\ \boldsymbol{CA}^{n-1} \end{bmatrix}, \qquad \text{(Equation S 10)}$$

is equal to n. For the exemplary state-space representation of a resistive-capacitive impedance function identified by ELSA comprising an ohmic resistance and RC elements in (Equation 21)-(Equation 23), we receive a SISO systems with $p = 1$ and $q = 1$, that is fully decoupled, i.e., with $\boldsymbol{A}$ being a diagonal matrix, and with $\boldsymbol{C} = [c_1 \quad \cdots \quad c_n] = [1 \quad \cdots \quad 1]$. If all time constants are different from each other, the observability matrix has the full rank n, i.e., all states are observable. The reverse conclusion is just as valid for ELSA, processes with the same (or really close) time constants cannot not be separated and will hence occur as one state variable.

Identifiability of parameters

The maximum number of structurally identifiable parameters N_{ip} of an LTI system with fully occupied matrices $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, and $\boldsymbol{D}$ is less or equal to the number of non-zero elements contained in the four system matrices:

$N_{\mathrm{ip}} \leq N_{\boldsymbol{A}} + N_{\boldsymbol{B}} + N_{\boldsymbol{C}} + N_{\boldsymbol{D}} = n^2 + n \cdot p + q \cdot n + q \cdot p\,.$ (Equation S 11)

If the system description contains more parameters than matrix elements, only lumped elements can be identified.

For the state-space representation of the resistive-capacitive example in (Equation 21)-(Equation 23), the maximum number of identifiable parameters is

$N_{\mathrm{ip}} \leq N_{\boldsymbol{A}} + N_{\boldsymbol{B}} + N_{\boldsymbol{C}} + N_{\boldsymbol{D}} = n + n + 0 + 1 = 2n + 1\,.$ (Equation S 12)

Conclusion: ELSA can be used to determine an upper bound for the maximum number of structurally identifiable parameters of an impedance spectrum can be determined using the identified minimal model order of the resulting well-defined LTI state-space representation.

Note S4: Case differentiation for summands of single poles

For the impedance function of single pole elements in equation (12) $Z(j\omega) = \frac{a}{1+j\omega\, b}$, the signs of the parameters a and b can be positive or negative. For all combinations, a case differentiation of single poles is shown in Figure S1

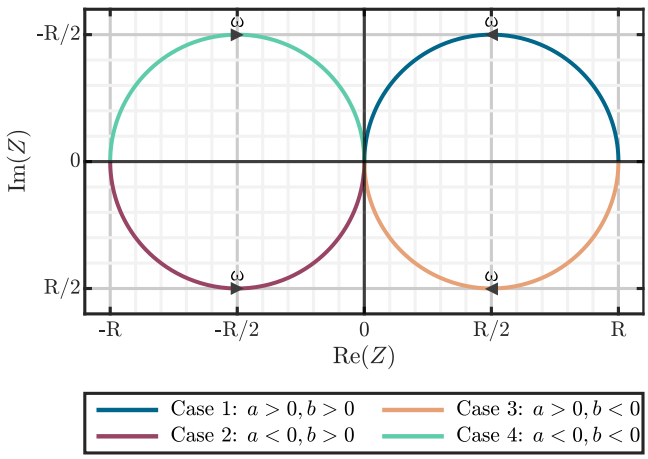


Figure S1. Case differentiation for summands of single poles.
Only for case 1 with positive values, $a > 0$ and $b > 0$, a physical interpretation of an RC circuit is immediately possible.

Case 2: $a < 0$ and $b > 0$

For negative a and positive b the transfer function of a PT1 element is converted into the transfer function of a DT1 element and a P element, i.e. of an RL element and a resistance:

$$h(j\omega) = \frac{-|a|}{1+j\omega\, b} + |a| - |a| = \frac{-|a|}{1+j\omega\, b} + \frac{|a|\cdot(1+j\omega\, b)}{1+j\omega\, b} - |a| = \frac{-|a| + |a| + j\omega\,|a|\, b}{1+j\omega\, b} - |a| =$$
$$= \frac{j\omega\,|a|\, b}{1+j\omega\, b} - |a| = -R + \frac{j\omega\, R\,\tau}{1+j\omega\,\tau} = Z_{\mathrm{R}} + Z_{\mathrm{RL}}(j\omega) \qquad \text{(Equation S 13)}$$

with the resistance of the RL element $R = |a|$, the time constant $\tau = L/R = b$ and an ohmic offset $R = -|a|$ that corrects the ohmic resistance in (Equation 18).

Case 3 and 4: $b < 0$

The appearance of negative time constants in impedance spectra is controversially discussed in the literature. Examples in the literature provide a kinetic interpretation of a negative time constant, such as for the dissolution/passive transition[45] or the glucose electrooxidation[46]. In both cases, we argue that the studied systems are not time-invariant during the impedance measurement. Therefore, we propose that non-stationarity or time-variance should be considered. Our observations of the occurrence of elements with negative time constants within the measured frequency range correlate with the occurrence of noise and distortion. Both effects are erased when the DRT is calculated via regularization. This implies that if we find elements with negative time constants using the ELSA method and cancel them in the transfer function, this has the same effect as regularization.

Note S5: Conversion of single pole elements with very high and very low time constant

The impedance function of a single pole element with separation into real and imaginary parts is

$$h(j\omega) = \frac{a}{1+j\omega\, b} = \frac{a\cdot(1-j\omega\, b)}{(1+j\omega\, b)\cdot(1-j\omega\, b)} = \frac{a}{1+\omega^2 b^2} - \frac{j\omega\, ab}{1+\omega^2 b^2} \qquad \text{(Equation S 14)}$$

with a and $b \in \mathbb{R}$ for the 4 cases discussed.

The Loewner method yields elements with time constants $\tau = |b|$ outside the measured frequency range. These can be converted to lumped serial elements of R, L, and C in accordance with the initial and final value theorem of the Laplace transform.

Time constants lower than the lower limit of the measurement range

For elements with very low time constants $\tau < (2\pi\, f_{\max})^{-1}$ the limit behavior shall be calculated. For any realistic sampling time, the step response already reached its final value. For $T \gg \tau$ the final value theorem applies, here used for comparison of coefficients and the Fourier transform:

$$h(j\omega) = \frac{a}{1+\omega^2 b^2} - j\omega\frac{ab}{1+\omega^2 b^2} \overset{\omega\to 0}{\approx} a - j\omega\, ab = Z_{\mathrm{R}} + Z_{\mathrm{L}}(j\omega) \qquad \text{(Equation S 15)}$$

with the limit value for the real part of the resistance

$$\lim_{\omega\to 0}\left\{\frac{a}{1+\omega^2 b^2}\right\} = a = R \qquad \text{(Equation S 16)}$$

and the limit value for the imaginary part for the inductance

$$\lim_{\omega\to 0}\left\{-\frac{ab}{1+\omega^2 b^2}\right\} = -ab = L\,. \qquad \text{(Equation S 17)}$$

Consequently, an element with a very low time constant $\tau < (2\pi\, f_{\max})^{-1}$ acts like an ohmic resistance with $R = a$ and an inductance with $L = -ab$.

Time constants larger than the upper limit of the measurement range

For elements with very large time constants $\tau > (2\pi\, f_{\min})^{-1}$ the limit behaviour shall be calculated. For $T \ll \tau$ the initial value theorem applies:

$$h(j\omega) = \frac{a}{1+j\omega\, b} = \frac{a}{1+\omega^2 b^2} - \frac{j\omega\, ab}{1+\omega^2 b^2} = \frac{a}{1+\omega^2 b^2} - \frac{j\omega}{j\omega}\cdot\frac{j\omega\, ab}{1+\omega^2 b^2} =$$
$$= \frac{a}{1+\omega^2 b^2} + \frac{1}{j\omega}\cdot\frac{\omega^2\, ab}{1+\omega^2 b^2} \overset{\omega\to\infty}{\approx} 0 + \frac{1}{j\omega}\cdot\frac{1}{b} = Z_{\mathrm{R}} + Z_{\mathrm{C}}(j\omega) \qquad \text{(Equation S 18)}$$

with the limit value for the real part

$$\lim_{\omega\to\infty}\left\{\frac{a}{1+\omega^2 b^2}\right\} = 0 \qquad \text{(Equation S 19)}$$

and the limit value for the imaginary part using l`Hôpital`s rule

$$\lim_{\omega\to\infty}\left\{\frac{\omega^2\, ab}{1+\omega^2 b^2}\right\} = \lim_{\omega\to\infty}\left\{\frac{2\omega\, ab}{2\omega\, b^2}\right\} = \frac{a}{b} = \frac{1}{C}\,. \qquad \text{(Equation S 20)}$$

Consequently, an element with a very large time constant $\tau > (2\pi\, f_{\min})^{-1}$ acts like a capacitance with $C = ba^{-1}$.

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Table S1. Limiting behavior of single pole functions

Condition	Limit behavior	Parameters	Reference
$\tau < (2\pi\, f_{\max})^{-1}$	$\lim_{\omega\to 0}\{h(j\omega)\} = a - j\omega\, ab = \\ = Z_{\mathrm{R}} + Z_{\mathrm{L}}(j\omega) = R + j\omega\, L$	$R = \mathrm{a}$, $L = -ab$	(Equation S 15)
$\tau > (2\pi\, f_{\min})^{-1}$	$\lim_{\omega\to\infty}\{h(j\omega)\} = \frac{1}{j\omega}\cdot\frac{a}{b} = \\ = Z_{\mathrm{C}}(j\omega) = (j\omega\, C)^{-1}$	$C = \mathrm{b}/a$	(Equation S 18)

Note S6: Additional results of the electrochemical system analysis of a lithium-ion battery

Figure S2 (a) shows the Bode plot of the measured and simulated spectra of a cylindrical LFP | graphite cell from A123 Systems. Figure S2 (b) depicts the zeros, poles, and gain of the according ELSA result.

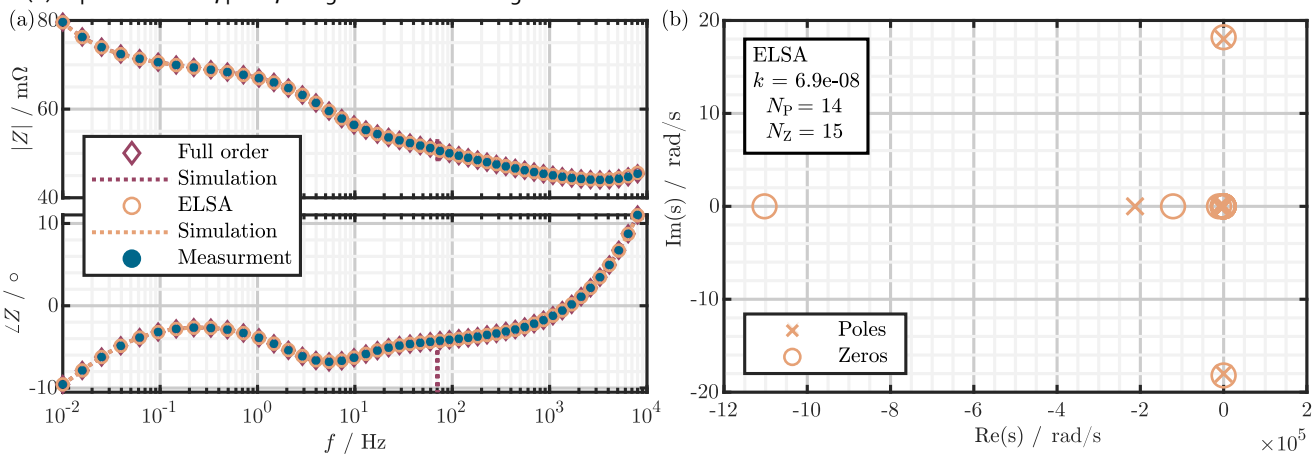


Figure S2. Bode plot and resulting zeros, poles, and gain

(a) Bode plot of the measured and simulated spectra of a cylindrical LFP | graphite cell from A123 Systems and (b) zeros, poles, and gain of the ELSA result.

Note S7: Electrochemical system analysis of further electrochemical technologies

The experimental details and references to data sources of four other test objects can be found in Table S2.

Table S2: Information on further test objects, operating conditions, measurement equipment and conditions as well as references.

	Lithium-ion battery	High-temperature polymer electrolyte fuel cell	Double-layer capacitor	Analog RLC circuit
Technology details	Panasonic NCR-18650B, 18650 nickel-manganese-cobalt-oxid | graphite, 2.5 Ah	Custom made fuel cell assembly, metallic bipolar plates with single serpentine flow-fields, electrode surface area of 4 cm^2, polybenzimidazole (PBI) membranes (Dapazol1, Danish Power Systems-DPS1)	Maxwell, capacitance 3400 F, 2.85 V	Custom made circuit with a parallel connection of a 2.2 µF foil capacitor and a lossy inductor with 9.7 mΩ and 5 mH, factory values may vary ± 10 %
Operating point	50 % state of charge, 3.27 V, 20 °C	H2 / air stoichiometry 1.8 / 2.5, 160 °C, 800 mA/cm^2	voltage 603.5 mV, temperature 30 °C	No specific operating point
Potentiostat / frequency	Zahner Zennium Pro workstation	Zahner Zennium Pro workstation	Zahner Zennium Pro workstation	Biologic VMP300

response analyzer				
EIS mode	galvanostatic	galvanostatic	potentiostatic	galvanostatic
Amplitude	250 mA	100 mA	10 mV	30 mA
Min. frequency	10 mHz	100 mHz	100 mHz	100 Hz
Max. frequency	10 kHz	100 kHz	1 kHz	10 kHz
Number of frequencies	49	58	29	81
Reference	[50]	[51]	[27]	-

Lithium-ion battery

For the analysis of the second lithium-ion battery, now with NMC | graphite, cell index 78 of the publicly available dataset [50] is used. It contains impedance measurement of 92 Panasonic NCR-18650B lithium-ion batteries at the begin of life. All measurements are validated through the extended Kramers-Kronig test according to [41].

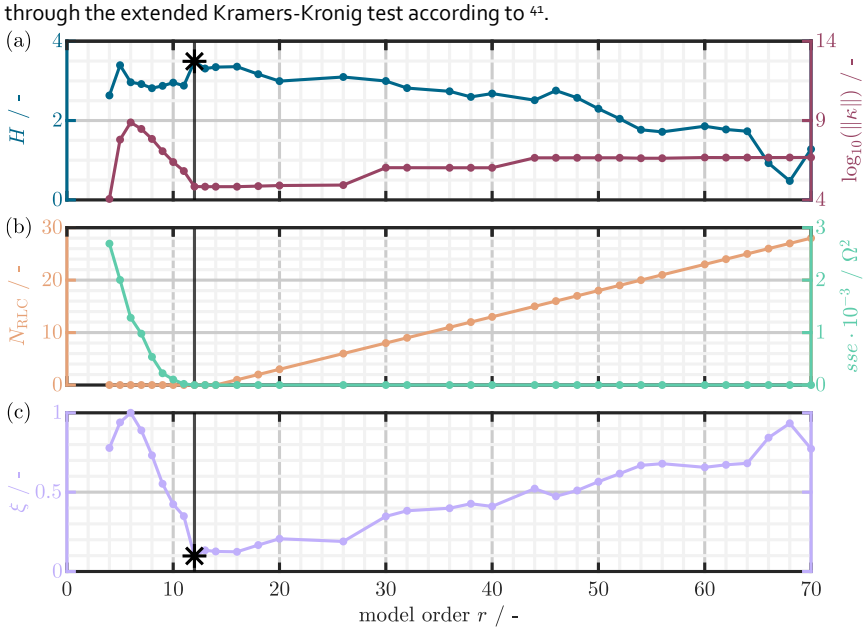


Figure S3. Selection of minimal model order of a lithium-ion cell

Selection of minimal model order of an NMC | graphite cell from Panasonic by (a) norm of curvature $\boldsymbol{\kappa}$, Shannon entropy $\boldsymbol{H}$, and c) search function $\boldsymbol{\xi}$. Sum of squared errors $\boldsymbol{sse}$ and number of RLC elements $\boldsymbol{N_{\mathrm{RLC}}}$ in (b).

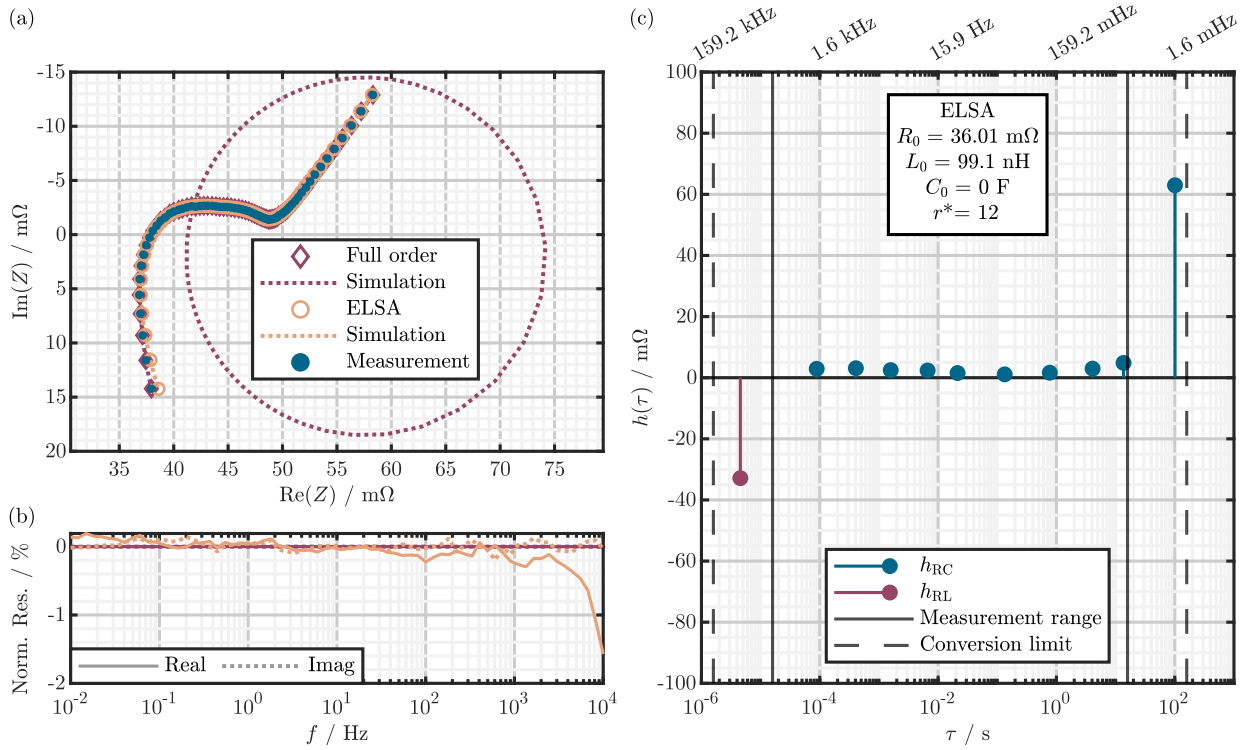


Figure S4. ELSA results of a lithium-ion cell

Analysis results of a cylindrical NMC | graphite cell from Panasonic: (a) Nyquist plot of measured and simulated impedance spectrum, (b) normalized residuals of the measured and reduced impedance spectra, and (c) distribution of polarization for RC and RL elements.

Simulation of the full-order model in **Figure S4**(a) presents a large impedance loop, a clear indication of overfitting. The identified minimal model order in **Figure S3** is 12. The distributed polarization $h(\tau)$ for RC and RL elements at their specific time constants are shown in **Figure S4**(c): a single RL element for the resistive-inductive high frequency tail and a distribution of RC elements for the resistive-capacitive impedance arch and the diffusive impedance tail. As neither RLC circuits nor single poles of cases 3 and 4 are found, the measured spectrum is fully described by non-resonant LTI elements with positive model parameters. Hence, the spectrum shows no hints of non-linearity nor of time-variance. An identified element of case 3 with a time constant more than a frequency decade outside the measured frequency range is converted into a serial ohmic resistance and an inductivity according to **Table S1**. The resulting values of R_0, L_0, and C_0 are given in **Figure S4**(c). Overall, the ELSA spectrum describes the measured impedance data with high accuracy without overfitting. The low normalized residuals in (b) show a slight increase in the real part for high frequencies that originates from the conversion of a case 3 element to an ohmic resistance and an inductivity according to (Equation S 15).

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High-temperature polymer electrolyte fuel cell

The impedance spectrum of a high-temperature polymer electrolyte fuel cell stems from Weiß et al. [51] is shown in **Figure S6**.

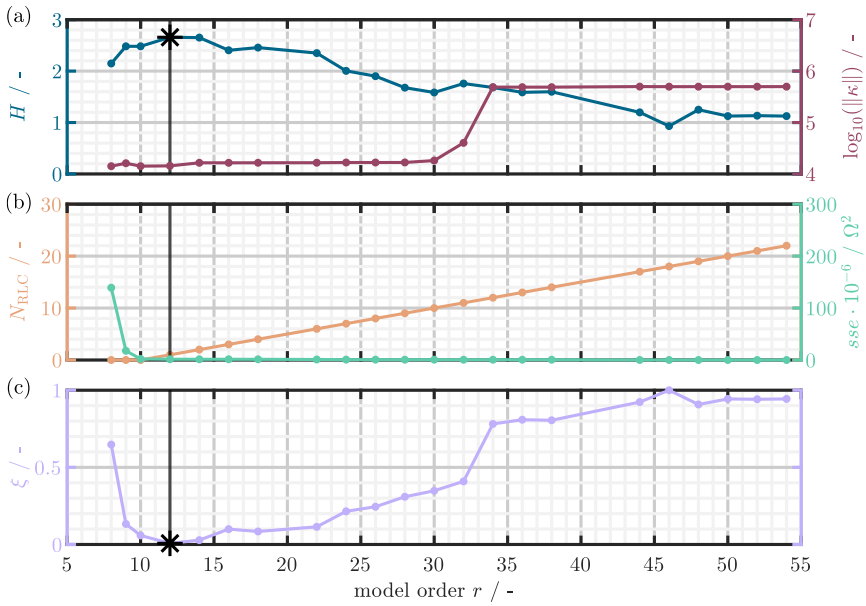


Figure S5. Selection of minimal model order for a high-temperature PEM fuel cell

Selection of minimal model order for the impedance spectrum of a high-temperature PEM fuel cell by (a) norm of curvature $\boldsymbol{\kappa}$, Shannon entropy $\boldsymbol{H}$, and (c) search function $\boldsymbol{\xi}$. Sum of squared errors $\boldsymbol{sse}$ and number of RLC elements $\boldsymbol{N_{RLC}}$ in (b).

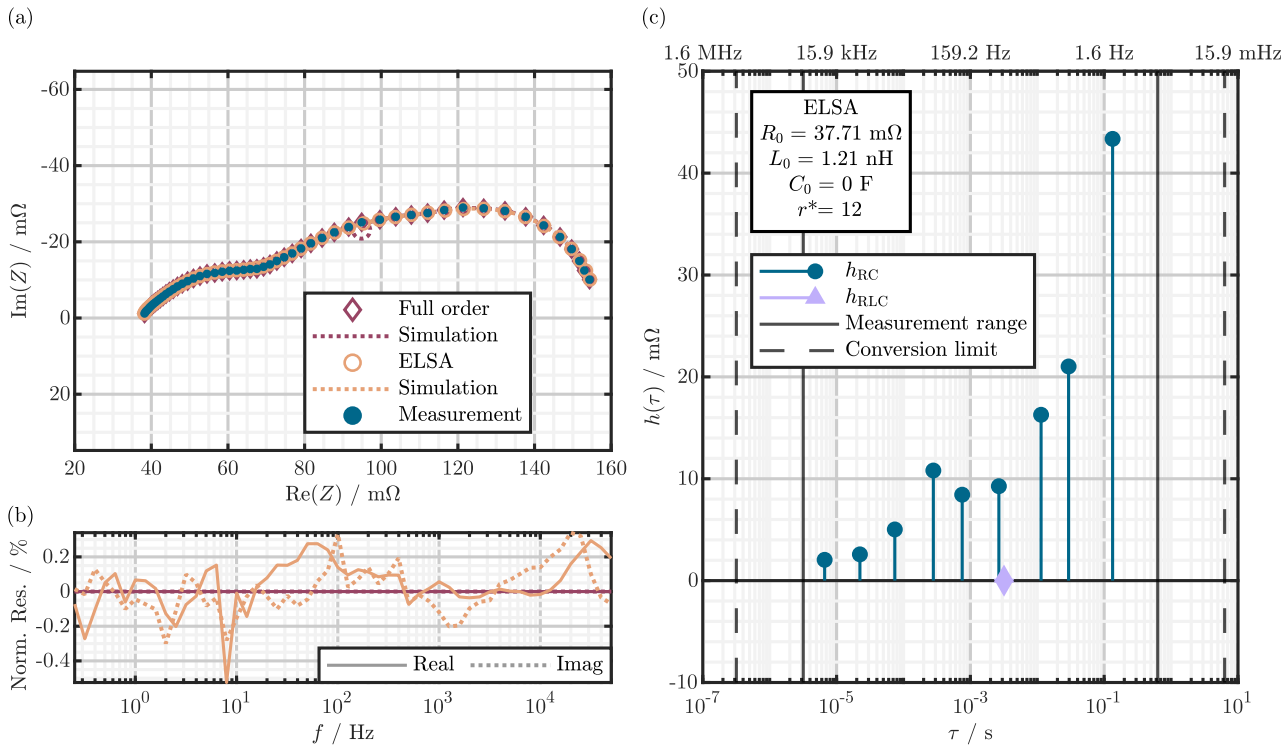


Figure S6. ELSA results of a high-temperature PEM fuel cell

Analysis results of a high-temperature PEM fuel cell: (a) Nyquist plot of measured and simulated impedance spectrum, (b) normalized residuals of the measured and reduced impedance spectra, and (c) distribution of polarization for RC and RLC elements.

Small impedance loops in Figure S6(a) indicate an overfitting of the full-order model. The identified minimal model order in Figure S5 is 12. The distributed polarization $h(\tau)$ for RC and RLC elements at their specific time constants are shown in Figure S6(c): a distribution of nine RC elements for the resistive-capacitive impedance arches and a single but small RLC element with a characteristic time constant close to the Dirac pulse of the sixth RC element. Hence, the spectrum shows only minor hints of non-linearity or time-variance. An identified element of case 3 with a time constant more than a frequency decade outside the measured frequency range is converted into a serial ohmic resistance and an inductivity according to Table S1. The resulting values of R_0, L_0, and C_0 are given in Figure S6(c). Overall, the ELSA model precisely describes the measured impedance data over the full frequency range without overfitting.

A comparison of the calculated DRT of gDRT and ELSA in Figure S7 shows a high level of agreement between both distribution functions, the Diracs of ELSA and the peaks of gDRT, and a significant difference at high time constants. The steep flank of the large peak in the gDRT brings about an artefact, the RL peak at high time constants, presumably an effect of the regularization.

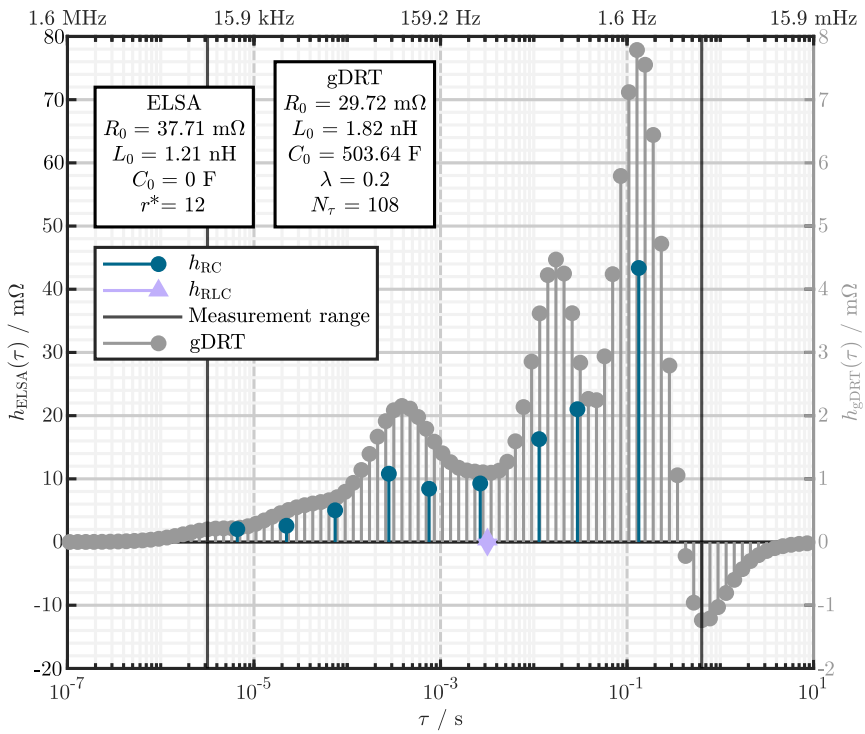


Figure S7. Comparison of distributed polarizations

Comparison of distributed polarization of a high-temperature polymer electrolyte fuel cell identified by gDRT and ELSA.

Double-layer capacitor

The impedance spectrum of a double-layer capacitor in **Figure S9**(a) was examined using the generalized DRT analysis in [27].

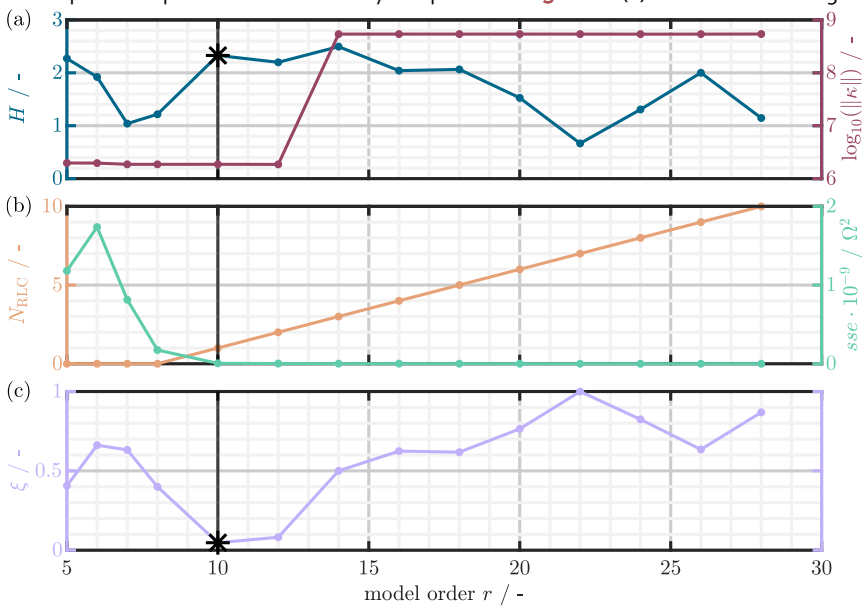


Figure S8. Selection of minimal model order for a double-layer capacitor

Selection of minimal model order for the impedance spectrum of a double-layer capacitor by (a) norm of curvature $\boldsymbol{\kappa}$ and Shannon entropy $\boldsymbol{H}$ and (c) search function $\boldsymbol{\xi}$. Sum of squared errors $\boldsymbol{sse}$ and number of RLC elements $\boldsymbol{N_{\mathrm{RLC}}}$ in (b).

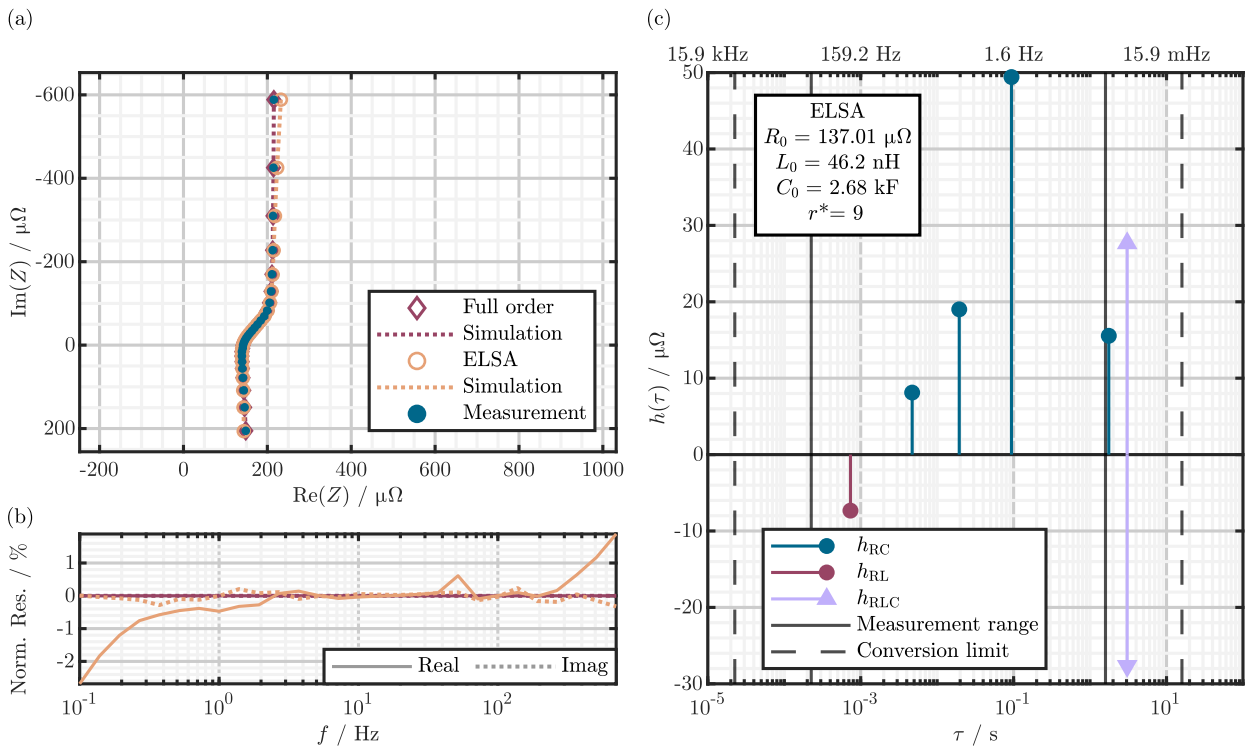


Figure S9. ELSA results of a double-layer capacitor

Analysis results of a double-layer capacitor (a) Nyquist plot of measured and simulated impedance spectrum, (b) normalized residuals of the measured and reduced impedance spectra, and (c) distribution of polarization for RC, RL, and RLC elements.

The identified minimal model order in **Figure S8** is 10. For the selection of minimal model order, it should be noted that for this spectrum the minimum in the search function ξ does not coincide with the global maximum of the Shannon entropy. Here, the global maximum of the Shannon entropy is in a range of model orders with higher norms of the curvature. The distributed polarization $h(\tau)$ for RC, RL, and RLC elements at their specific time constants are shown in **Figure S9**(c): a distribution of 4 RC elements for the resistive-capacitive impedance arch and a single RLC element with a large polarization contribution and a large characteristic time constant. As discussed for the redox flow battery, gDRT identifies an RC and an RL element instead of the RLC element in ELSA. For time constants below the conversion range, an element of case 2 and case 3 each is found and gets converted into an ohmic resistance and an inductivity. For time constants above the conversion range, an element of case 4 gets converted into a serial capacitance according to **Table S1**. The resulting values of R_0, L_0, and C_0 match precisely the results of the gDRT and are given in **Figure S9**(c). Overall, the ELSA model describes the measured impedance data with high accuracy over a wide frequency range with slightly diverging residuals in the real part for high and low frequencies.

Analog RLC circuit

To demonstrate the applicability of ELSA beyond electrochemical systems, and to validate its performance, an impedance spectrum of an analog RLC circuit is analyzed. The system is composed of real but linear components: L and C with given parameters including the measured ohmic resistance of the inductance in **Table S3**. The first part of the validation is to determine a model of a resonating system by its impedance data. The second part of the validation is the comparison of the identified and the given parameters.

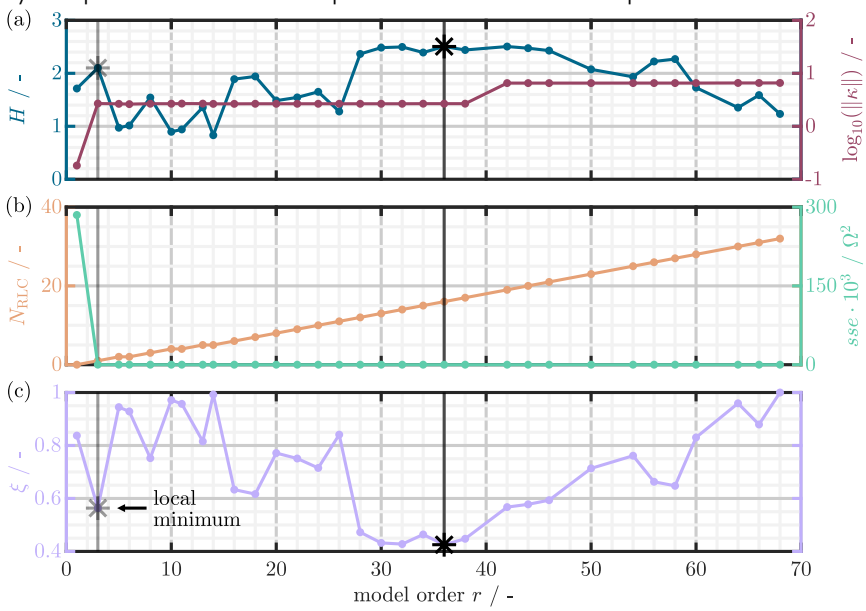


Figure S10. Selection of minimal model order for an analog RLC circuit

Selection of minimal model order for the impedance spectrum of an analog RLC circuit by (a) norm of curvature $\boldsymbol{\kappa}$ and Shannon entropy $\boldsymbol{H}$ and (c) search function $\boldsymbol{\xi}$. Sum of squared errors $\boldsymbol{sse}$ and number of RLC elements $\boldsymbol{N_{RLC}}$ in (b).

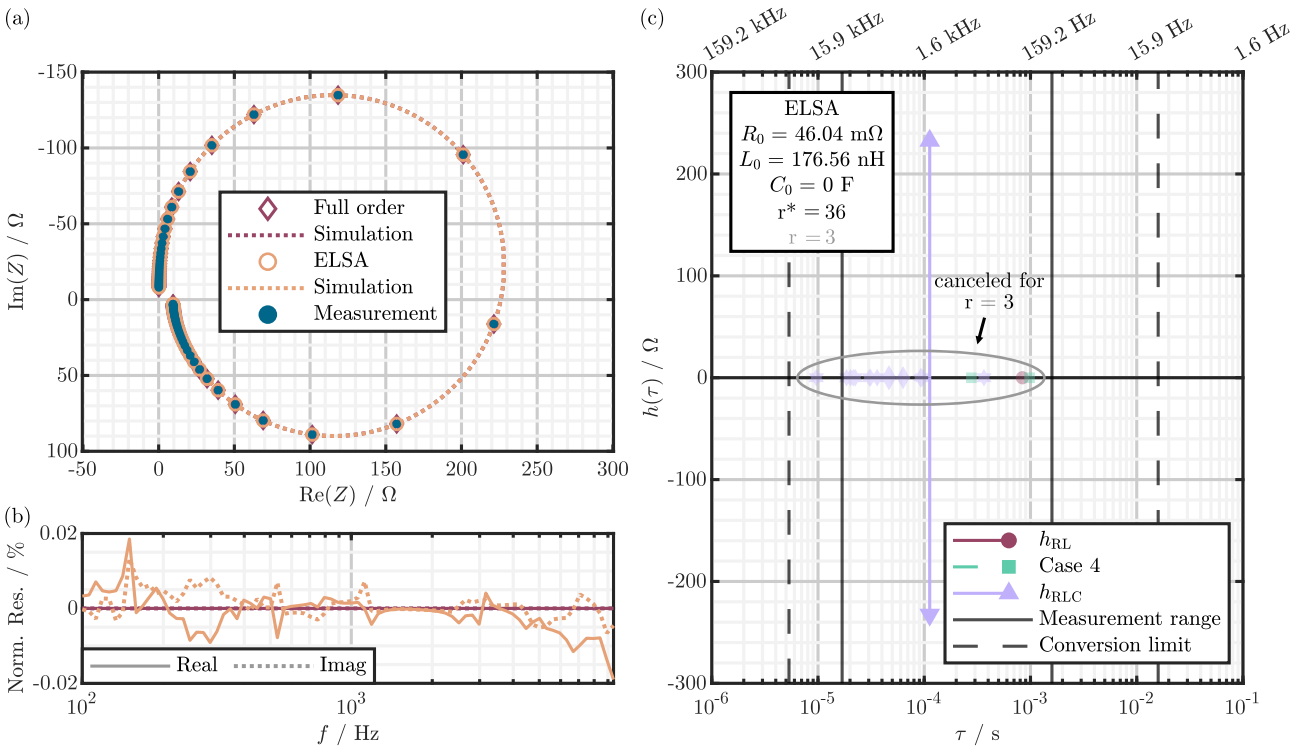


Figure S11. ELSA results of an analog RLC circuit

Analysis results of an analog RLC circuit: (a) Nyquist plot of measured and simulated impedance spectrum, (b) normalized residuals of the measured and reduced impedance spectra, and (c) distribution of polarization for RL, RLC, and case 4 elements.

The full-order spectrum in **Figure S11**(a) shows no signs of overfitting for the full-order model. The selection of the model order in **Figure S10** shows a global minimum for the search function ξ at a high model order of $r^* = 36$ and a local minimum at $r = 3$. The found minimum at a high number of RLC elements is plausible if we regard the low impedance (few mΩ) at low and high frequencies where the potentiostat works at low measurement accuracies. Nonetheless, ELSA proves capable of determining a model for this resonating system with extremely high accuracy over the full frequency range based on its impedance data. Other DRT tools that have been tested using this spectrum fail to find an appropriate model description. Additionally, ELSA precisely identifies the parameters of the utilized circuit components as shown in the comparison to the given parameters of **Table S3**. For a complementary validation of the values of the electronic components, provided by the manufacturer and identified by ELSA, an equivalent circuit model, consisting of a serial resistance and inductance in parallel to a capacitance, was fitted to the measured data.

Table S3. Given and identified parameters for analog RLC circuit elements.

Element	Electronic component value	ECM fit ([R-L] || C)	ELSA ($r^* = 36$)	ELSA ($r = 3$)
Inductor	$R_L \approx 9.7\ \Omega$ (*)	9.67 Ω	8.49 Ω	9.55 Ω
	$L = 5\ \text{mH} \pm 1\ \%$	4.83 mH	4.95 mH	4.9 mH
Capacitor	$R_C \to 0\ \Omega$ (Δ)	–	–	–
	$C = 2.2\ \mu\text{F} \pm 10\ \%$	2.20 µF	2.55 µF	2.27 µF

(Δ) Considered to be negligibly small
(*) Measured with a multimeter